

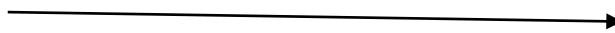
SAE INDIA

SOUTHERN SECTION

Bicycle Design Challenge 2025-26



Design Report - 2



Team Name	Pegasus
Team ID	BDC2025020

NAME	TEAM COMPOSITION
Bhuvanan M	Captain
Kusumarekha M	Vice captain
Dharaneshwaran B	Designer-1
Ambika B	Designer-2
Aravinth R	Driver
Keerthana M	Co-Driver
Dharanidharan T	Research and development

FACULTY MENTOR:

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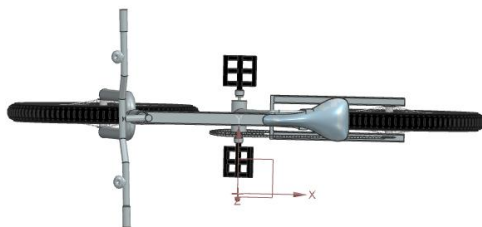
CONTENTS:

- **CAE/FEA-Reports**
- **Design Modification Based on Validation Report**
- **Part drawings 2 after CAE/FEA**

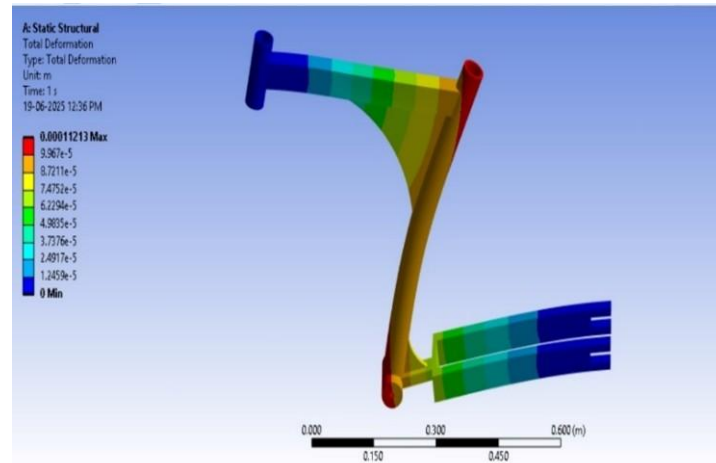
OVERVIEW:

Our Z – frame bicycle Address

The bicycle features a unique Z-frame with a minimalist structure, including front and rear wheels of equal size, a standard chain drive mechanism, and a crank set with pedals. The saddle is mounted on an inclined seat post, and the handlebars are angled for ergonomic use. A local coordinate system near the crank indicates the model used for kinematic or structural analysis.

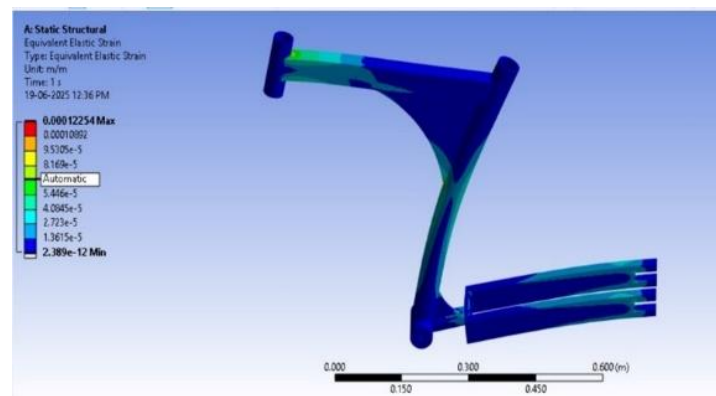


FEA (FINITE ELEMENT ANALYSIS) SECTION OF THE REPORT



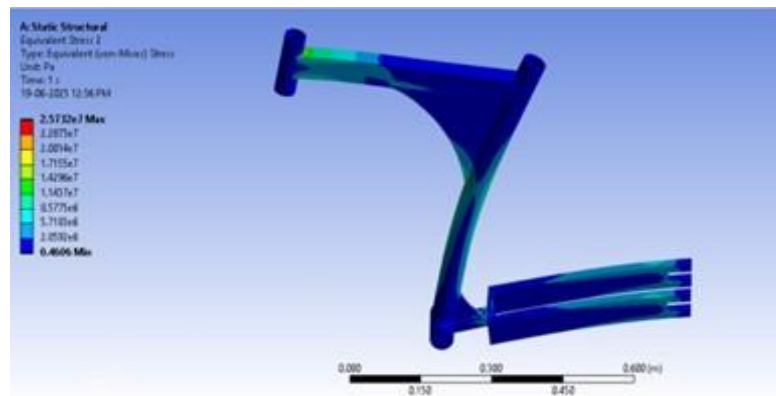
1.TOTAL DEFORMATION

The test is called static structural analysis and it checks the total deformation (how much the frame moves). Red color shows the area that bends the most. Blue color shows the area that moves the least. The maximum bending is very small, around 0.1 mm, which means the frame is strong.



2.EQUIVALENT STRAIN

The analysis type is called Equivalent Elastic Strain . It tells how the material inside the frame reacts under load. Blue areas show very little or no strain. The highest strain is very small (0.00012254), which means the material is not overstressed. This test helps check if the frame will stay strong and not get damaged while riding.



3.EQUIVALENT STRESS

Red areas show the highest stress, and blue areas show the lowest. The maximum stress is around 2.57×10^7 Pa .Most of the frame shows low stress (blue and green colors), meaning it's safe and strong. This test helps confirm the frame can handle real-life forces without cracking or breaking.

4.OVERALL CONCLUSION

The bicycle frame design is mechanically safe, strong, and stable. All results show that the frame:

- Experiences low deformation
- Handles minimal internal strain
- Shear stress well within safe limits

This proves the frame is suitable for real-world use and can withstand typical cycling loads without damage.

DESIGN MODIFICATIONS BASED ON VALIDATION RESULTS

Based on the results of total deformation, equivalent strain, and equivalent (von-Mises) stress analysis, the bicycle frame performs well overall. However, some design improvements can enhance strength, durability, and safety even further:

◇ 1. Reinforce High-Stress Areas :

- **Observation:** Maximum stress and deformation occur near joints (e.g., seat post joint, bottom bracket).
 - **Modification:**
 - Add fillets at stress-concentrated joints to better distribute loads.
 - Use thicker tubing at these joints.
-

◇ 2. Optimize Tube Thickness :

- **Observation:** Most areas show low stress and strain, suggesting over-design.
 - **Modification:**
 - We are Reduce material thickness in low-stress areas (like chain stays or top tube) to save weight.
 - Maintain or increase thickness only in high-stress zones
-

◇ 3. Improve Frame Geometry:

- **Observation:** The Z-frame shape causes stress to concentrate in specific curved areas.
 - **Modification:**
 - Slightly adjust the angles or curvatures to improve load transfer.
 - Ensure load paths flow more directly from seat and handlebar to the crank area.
-

◇ 4. Material Consideration :

- **Observation:** Stress is within safe limits, but some strain concentration suggests that the stiffness of the frame could be improved to enhance performance under load.
 - **Modification:**
 - Use Cold Rolled Steel (CRS) as the primary frame material to improve strength and stiffness. CRS offers higher yield strength and better fatigue resistance compared to basic steel.
 - CRS provides a good balance between cost, strength, and manufacturability, making it suitable for high-load areas of the frame.
 - Ensure protective coatings (e.g., paint or powder coating) are applied to prevent corrosion, as CRS is not naturally rust-resistant.
-

◇ 5. Safety Factor Adjustment :

- **Observation:** While safe under current load, real-world conditions include impacts and fatigue.
 - **Modification:**
 - Design with higher safety factors (e.g., 2.0 or above) especially in the crank and seat joint areas.
 - Conduct fatigue analysis if repetitive loading is expected.
-

◇ Modifications for Enhanced Ergonomics & Usability:

- **Z-Frame Structure:**

The unique Z-shaped frame provides a modern look while supporting better load distribution and rider balance.
- **Seat-Mounted Spring Suspension:**

A suspension system below the seat absorbs road shocks, improving comfort on uneven terrain and reducing rider fatigue.
- **CR (Cold Rolled) Steel Frame Material:**

Provides excellent strength, stiffness, and durability with a smooth surface finish for an improved riding experience.
- **Adjustable Seat Post:**

Allows Horizontal length adjustment to fit riders of different sizes and ensure proper leg extension while pedaling.
- **Ergonomic Handlebar with Enhanced Grip:**

Designed for natural wrist alignment and includes a non-slip grip to improve control and reduce hand fatigue.

- **Pivoting Handlebar Stem:**

Enables adjustment of reach and angle, helping riders maintain an upright posture or lean-forward stance as preferred.

- **Platform Pedals:**

Wide surface area supports a stable foot grip and is suitable for all footwear types, enhancing safety.

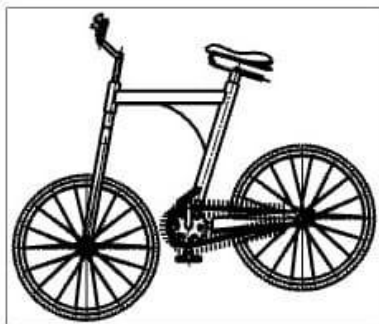
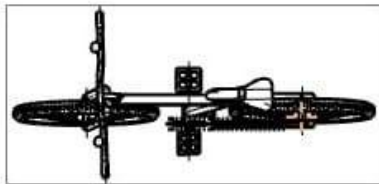
- **Side-Mounted Kickstand:**

Easy to deploy and supports the bike upright on various surfaces, adding convenience for parking.

- **Eco-Friendly Components:**

Use of low-emission coatings aligns with sustainable design principles.

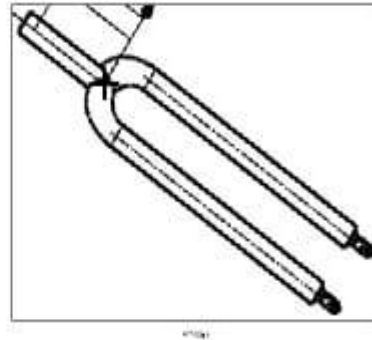
PART DRAWINGS -2 AFTER (FEA)



Not for sale - Not to be used for commercial purposes

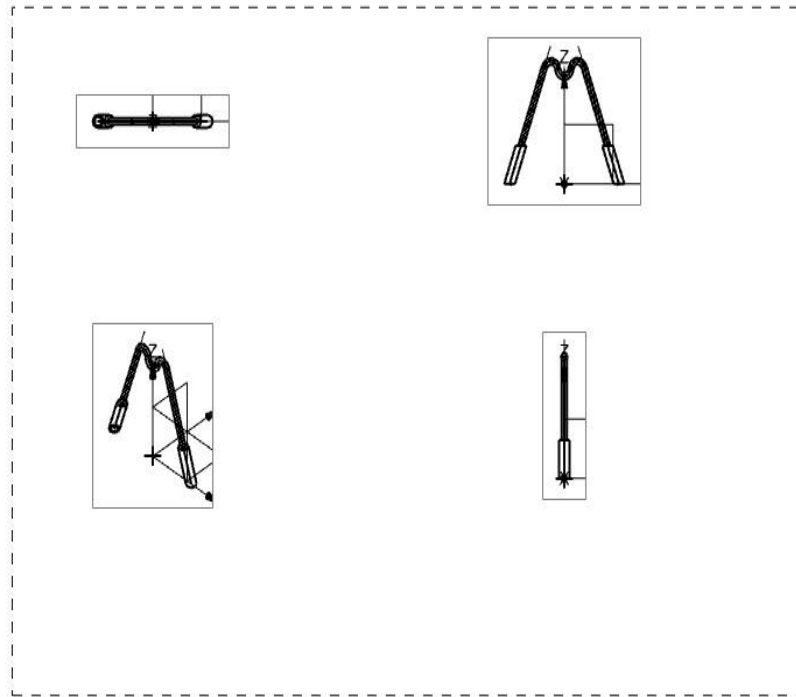


THE FRONT FORK OF A BICYCLE



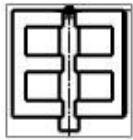
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Team Name	Pegasus
College	Knowledge Institute of Technology

SIDE-MOUNTED KICKSTAND:



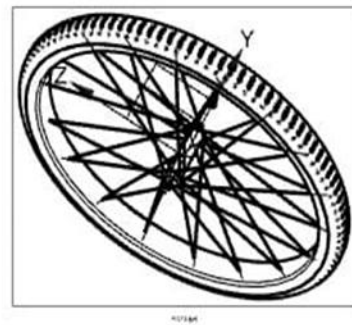
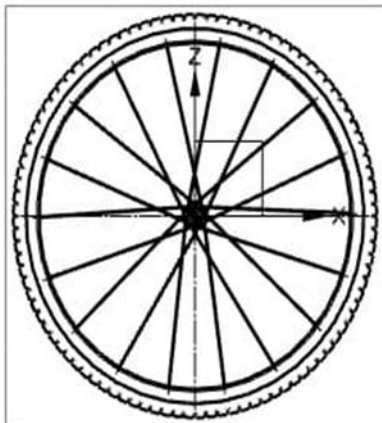
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PEDAL OF A BICYCLE



Scale 1:1	ALL DIMENSIONS ARE IN mm
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TYRE OF A BICYCLE



Scale 1:1	ALL DIMENSIONS ARE IN mm
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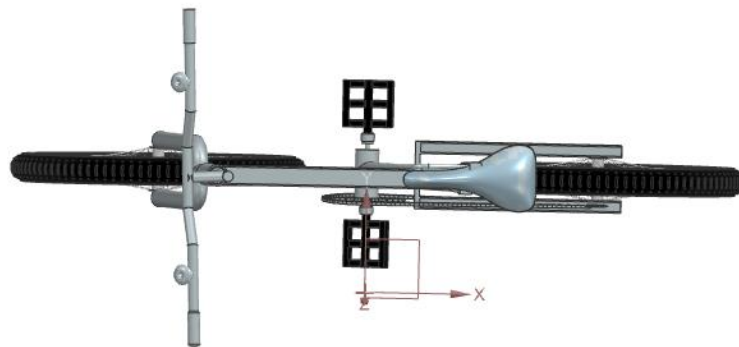
3D VIEWS:



SIDE VIEW



FRONT VIEW



TOP VIEW

CONCLUSION

Our team, **Pegasus**, has developed a practical, stable, and bicycle featuring a Z-frame and a spring-based seat suspension for enhanced comforts. Our design ensures durability, user convenience, and aligns with modern urban commuting needs.

thank you 